Zi Huang

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Education

Delft University of Technology (TU Delft) Sep 2024 — Present
Master of Science in Robotics *Delft, NL*
- CGPA: 8.5/10
- Key Coursework: Deep Reinforcement Learning, Robot Dynamics, Planning and Decision Making, Machine Perception, Computer Vision, Machine Learning.
- Thesis: Self-Adaptive Spiking Reservoir for Reinforcement Learning (Supervisor: Dr. Cosimo Della Santina)

Harbin Institute of Technology (HIT) Sep 2020 — Jun 2024
Bachelor of Engineering in Automation *Weihai, CHN*
- CGPA: 91.47/100
- Key Coursework: Modern Control Theory, Intelligent Control System, Embedded Systems, Calculus & Linear Algebra.

Professional Experience

Software Engineer (AI & Computer Vision) Jul 2025 — Nov 2025
Havatec BV *Nieuw-Vennep, NL*
- Developed an end-to-end computer vision pipeline for instance segmentation.
- Engineered the complete life cycle: curated custom datasets, trained Deep Learning models in PyTorch, and optimized inference using ONNX for real-time deployment.
- Integrated C# post-processing logic to interface AI predictions with industrial machine control systems; the segmentation model was subsequently adopted by a colleague for robotic flower picking from cluttered piles.

Mechanical Engineer (Control) Jun 2023 — Sep 2023
Chengdu RIG Science & Technology Co., Ltd *Chengdu, CHN*
- Designed and tuned cascaded PID algorithms for precise motor control (speed, position, and torque).
- Validated control stability and performance through extensive MATLAB/Simulink simulations.

Publications

[1] **Z. Huang**, C. Zhang, W. Pan, and C. Della Santina, "Hierarchical Reservoir Control for Soft Snake Robot Locomotion," in *Workshop on Nonlinear Physical Intelligence in Soft Machines, IEEE RAS Int. Conf. on Soft Robotics (RoboSoft)*, Kanazawa, Japan, Apr. 2026.

Research & Academic Projects

Diffusion-Policy Based Deformable Manipulation Feb 2026 — Present
- Advisors: Dr. Cosimo Della Santina, Zhaoting Li
- Developed a dual-arm robotic system for garment folding as part of the ICRA 2026 (WBCD Competition).
- Designed an interactive imitation learning framework that incorporates both human corrective actions (positive samples) and policy-generated failures (negative samples) for contrastive diffusion policy training.
- Improved sample efficiency and generalization compared to behavior cloning, reducing overfitting while enhancing task performance.

Master's Thesis: Self-Adaptive Spiking Reservoir for Reinforcement Learning Oct 2025 — Present
- Supervisor: Dr. Cosimo Della Santina, Chuhan Zhang
- Formulated a novel contrastive and auto-encoding training methodology for reservoir-computing networks, surpassing randomly generated reservoirs across diverse tasks, seamlessly integrated with Proximal Policy Optimization (PPO).
- Achieving reduced power consumption through the deployment of Spiking Neural Networks (SNNs).
- Validated the framework through extensive experiments spanning Gymnasium control benchmarks, simulated soft snake robot locomotion, and on a physical Franka manipulator for dynamical reaching tasks.

Apple Harvesting Robot Manipulation Apr 2025 — Jun 2025
- Video: https://youtu.be/dJFtMQb1bO0
- Architected a distributed ROS2 network integrating LiDAR, ultrasound, RGBD camera, and joint encoders across asynchronous nodes for robust navigation, perception, and control.
- Designed and implemented a whole-body motion planning pipeline combining A* search for mobile base navigation with MoveIt for arm trajectory generation, achieving coordinated pick-and-place harvesting through decomposed base-arm motion.

Combinatorial Generalization in VAEs Feb 2025 — Apr 2025

- Code: https://github.com/Roodster/dsait4125-cv
- Investigated representation learning by modeling group actions in Variational Autoencoders (VAEs), a core challenge for generalizable visual reasoning in robotic perception.
- Visualized latent space disentanglement using Uniform Manifold Approximation and Projection (UMAP) to compare transformation-encoding VAEs against standard baselines.

Multi-Sensor Fusion for 3D Perception Nov 2024 — Jan 2025

- Developed a robust cross-modal sensor fusion pipeline integrating LiDAR point clouds (bounding box proposals), camera images (pedestrian classification), and Radar data (velocity-based adaptive confidence scaling).
- Achieved a 0.368 mAP on the "View of Delft" autonomous driving dataset, outperforming baseline models by 26%.

Safe Planning & Control for Quadrotors Nov 2024 — Jan 2025

- Advisor: Prof. Dr. Javier Alonso-Mora
- Video: https://youtu.be/btgGNvN8GC0
- Developed a hybrid planning framework in PyBullet combining Informed RRT* for asymptotically optimal global path search with a linear Model Predictive Controller (MPC) via the OSQP solver for precise trajectory tracking under dynamic constraints.

Bachelor's Thesis: Visual Navigation for Automated Guided Vehicle Nov 2023 — Jun 2024

- Video: https://youtu.be/fwzMp7Wbgtw
- Designed a robust RGB-D perception system utilizing Region Growing algorithms for accurate linear feature extraction in unknown environments.
- Integrated an Artificial Potential Field (APF) local path planner to enable real-time obstacle avoidance on a physical Automated Guided Vehicle (AGV).
- Validated the navigation stack's sim-to-real transferability through comprehensive MATLAB simulations and successful hardware deployments.

Honors & Awards

Outstanding Graduate, Dept. of HR & Social Security of Shandong Province Jan 2024

- Top 5% in the province.

Finalist, Interdisciplinary Contest in Modeling (ICM), COMAP Jul 2023

- Top 2% globally.

Academic Excellence Scholarships (First & Second Grade), Harbin Institute of Technology 2021 - 2023

- Awarded 5 times consecutively for maintaining outstanding GPA.
- consistently ranked in the Top 3% (First Grade) and Top 7% (Second Grade) of the cohort.

Second Prize, National Ocean Navigation Device Competition, Chinese Society of Naval Architects and Marine Engineers Aug 2021

- Designed and built an underwater robot capable of autonomous obstacle detection.

Outstanding Student Leader & Social Work Scholarship, Harbin Institute of Technology 2021 - 2022

- Awarded 3 times for exceptional contribution to the University Admission Office and volunteer services.

Teaching & Leadership

Teaching Assistant: Machine Perception Nov 2025 — Jan 2026
TU Delft *Delft, NL*

- Assisting in material preparation and practical sessions for graduate-level Machine Perception course.

Volunteer Team Lead Mar 2022 — Aug 2022
Chengdu Weiguang Public Welfare *Panzhihua, CHN*

- Led a team of 9 volunteer teachers to provide education in under-resourced areas.

Skills

- **Programming:** Python, C/C++, C#, Matlab, Simulink.
- **Robotics & AI:** Isaac Lab, Isaac Sim, ROS2, PyTorch, MoveIt, OpenCV, Git, Docker.
- **Planning & Logic:** PDDL, Prolog.
- **Methods:** Sim-to-Real Transfer, Deep Reinforcement Learning, Vision-Language-Action Models.
- **Languages:** English (C1), Chinese (Native).